

Netflix data Analyst

August 1, 2026

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df = pd.read_csv('mymoviedb.csv', lineterminator = '\n')
```

```
[3]: df.head()
```

```
[3]:  Release_Date      Title \
0   2021-12-15  Spider-Man: No Way Home
1   2022-03-01           The Batman
2   2022-02-25           No Exit
3   2021-11-24           Encanto
4   2021-12-22  The King's Man
```

```
Overview  Popularity  Vote_Count \
0  Peter Parker is unmasked and no longer able to...  5083.954      8940
1  In his second year of fighting crime, Batman u...  3827.658      1151
2  Stranded at a rest stop in the mountains durin...  2618.087       122
3  The tale of an extraordinary family, the Madri...  2402.201      5076
4  As a collection of history's worst tyrants and...  1895.511      1793
```

```
Vote_Average  Original_Language      Genre \
0           8.3                en  Action, Adventure, Science Fiction
1           8.1                en      Crime, Mystery, Thriller
2           6.3                en      Thriller
3           7.7                en  Animation, Comedy, Family, Fantasy
4           7.0                en  Action, Adventure, Thriller, War
```

```
Poster_Url
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2  https://image.tmdb.org/t/p/original/vDHsLn0WKl...
3  https://image.tmdb.org/t/p/original/4j0PNHkMr5...
4  https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...
```

```
[4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   object
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url           9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
[5]: df['Genre'].head()
```

```
[5]: 0    Action, Adventure, Science Fiction
1           Crime, Mystery, Thriller
2                      Thriller
3    Animation, Comedy, Family, Fantasy
4    Action, Adventure, Thriller, War
Name: Genre, dtype: object
```

```
[6]: df.duplicated().sum()
```

```
[6]: 0
```

```
[7]: df.describe()
```

```
[7]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

```
[8]: # UP Coming Work Summary
# we have a dataframe consisting of 9827 rows and 9 columns.
# our dataset looks a bit tidy with no NaNs nor duplicated values.
```

```
# Release Date column needs to be casted into date time and to extract only the
    ↳ year value.
# Overview, Original_Language and Poster_Url wouldn't be so useful during
    ↳ analysis, so we'll drop them.
# there is noticable outliers in Popularity column
# Vote_Average better be categorised for proper analysis.
# Genre column has comma saperated values and white spaces that needs to be
    ↳ handled and casted into category.
```

```
[9]: df.head()
```

```
[9]:   Release_Date      Title \
0   2021-12-15  Spider-Man: No Way Home
1   2022-03-01      The Batman
2   2022-02-25      No Exit
3   2021-11-24      Encanto
4   2021-12-22  The King's Man

      Overview  Popularity  Vote_Count \
0  Peter Parker is unmasked and no longer able to...   5083.954      8940
1  In his second year of fighting crime, Batman u...   3827.658      1151
2  Stranded at a rest stop in the mountains durin...   2618.087       122
3  The tale of an extraordinary family, the Madri...   2402.201      5076
4  As a collection of history's worst tyrants and...   1895.511      1793

      Vote_Average  Original_Language      Genre \
0           8.3          en  Action, Adventure, Science Fiction
1           8.1          en      Crime, Mystery, Thriller
2           6.3          en      Thriller
3           7.7          en  Animation, Comedy, Family, Fantasy
4           7.0          en  Action, Adventure, Thriller, War

      Poster_Url
0  https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2  https://image.tmdb.org/t/p/original/vDHsLnOWK1...
3  https://image.tmdb.org/t/p/original/4jOPNHkMr5...
4  https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...
```

```
[10]: df['Release_Date'] = pd.to_datetime(df['Release_Date'])

print(df['Release_Date'].dtypes)
```

```
datetime64[ns]
```

```
[11]: df['Release_Date'] = df['Release_Date'].dt.year
```

```
df['Release_Date'].dtypes
```

```
[11]: dtype('int32')
```

```
[12]: df.head()
```

```
[12]:
```

	Release_Date	Title \	Overview	Popularity	Vote_Count \
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793

	Vote_Average	Original_Language	Genre \
0	8.3	en	Action, Adventure, Science Fiction
1	8.1	en	Crime, Mystery, Thriller
2	6.3	en	Thriller
3	7.7	en	Animation, Comedy, Family, Fantasy
4	7.0	en	Action, Adventure, Thriller, War

	Poster_Url
0	https://image.tmdb.org/t/p/original/1g0dhYtq4i...
1	https://image.tmdb.org/t/p/original/74xTEgt7R3...
2	https://image.tmdb.org/t/p/original/vDHsLnOWKl...
3	https://image.tmdb.org/t/p/original/4j0PNHkMr5...
4	https://image.tmdb.org/t/p/original/aq4Pwv5Xeu...

```
[13]: cols = ['Overview', 'Original_Language', 'Poster_Url']
```

```
[14]: df.drop(cols, axis = 1, inplace = True)
df.columns
```

```
[14]: Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
        'Genre'],
        dtype='object')
```

```
[15]: df.head()
```

```
[15]:
```

	Release_Date	Title	Popularity	Vote_Count \
0	2021	Spider-Man: No Way Home	5083.954	8940

1	2022	The Batman	3827.658	1151
2	2022	No Exit	2618.087	122
3	2021	Encanto	2402.201	5076
4	2021	The King's Man	1895.511	1793

	Vote_Average	Genre
0	8.3	Action, Adventure, Science Fiction
1	8.1	Crime, Mystery, Thriller
2	6.3	Thriller
3	7.7	Animation, Comedy, Family, Fantasy
4	7.0	Action, Adventure, Thriller, War

Categoring Votes_Average column

```
[16]: def catigorize_col(df, col, labels):

    edges = [df[col].describe()['min'],
              df[col].describe()['25%'],
              df[col].describe()['50%'],
              df[col].describe()['75%'],
              df[col].describe()['max']]

    df[col] = pd.cut(df[col], edges, labels = labels, duplicates = 'drop')
    return df
```

```
[17]: labels = ['Not_popular', 'below_avg', 'average', 'popular']

catigorize_col(df, 'Vote_Average', labels)
df['Vote_Average'].unique()
```

```
[17]: ['popular', 'below_avg', 'average', 'Not_popular', NaN]
Categories (4, object): ['Not_popular' < 'below_avg' < 'average' < 'popular']
```

```
[18]: df.head()
```

```
[18]: Release_Date      Title  Popularity  Vote_Count  Vote_Average \
0      2021  Spider-Man: No Way Home    5083.954      8940      popular
1      2022      The Batman    3827.658      1151      popular
2      2022      No Exit    2618.087      122    below_avg
3      2021      Encanto    2402.201     5076      popular
4      2021  The King's Man    1895.511     1793      average

      Genre
0  Action, Adventure, Science Fiction
1      Crime, Mystery, Thriller
```

```

2                Thriller
3 Animation, Comedy, Family, Fantasy
4 Action, Adventure, Thriller, War

```

```
[19]: df['Vote_Average'].value_counts() # for this syntax you can check which
      ↪ number of movie get which types category.
```

```
[19]: Vote_Average
Not_popular    2467
popular        2450
average        2412
below_avg      2398
Name: count, dtype: int64
```

```
[20]: df.dropna(inplace = True)

df.isna().sum()
```

```
[20]: Release_Date    0
Title                0
Popularity           0
Vote_Count           0
Vote_Average        0
Genre                0
dtype: int64
```

```
[21]: df['Genre'] = df['Genre'].str.split(', ')

df = df.explode('Genre').reset_index(drop=True)
df.head()
```

```
[21]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average \
0	2021	Spider-Man: No Way Home	5083.954	8940	popular
1	2021	Spider-Man: No Way Home	5083.954	8940	popular
2	2021	Spider-Man: No Way Home	5083.954	8940	popular
3	2022	The Batman	3827.658	1151	popular
4	2022	The Batman	3827.658	1151	popular

	Genre
0	Action
1	Adventure
2	Science Fiction
3	Crime
4	Mystery

```
[22]: df['Genre'] = df['Genre'].astype('category')
```

```
df['Genre'].dtypes
```

```
[22]: CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy',
                                'Crime',
                                'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                                'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                                'TV Movie', 'Thriller', 'War', 'Western'],
                                , ordered=False, categories_dtype=object)
```

```
[24]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Release_Date    25552 non-null  int32
 1   Title           25552 non-null  object
 2   Popularity      25552 non-null  float64
 3   Vote_Count      25552 non-null  int64
 4   Vote_Average    25552 non-null  category
 5   Genre           25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

```
[25]: df.nunique()
```

```
[25]: Release_Date    100
      Title          9415
      Popularity     8088
      Vote_Count     3265
      Vote_Average     4
      Genre          19
      dtype: int64
```

```
[26]: df.head()
```

```
[26]:   Release_Date      Title  Popularity  Vote_Count  Vote_Average  \
0      2021  Spider-Man: No Way Home    5083.954      8940    popular
1      2021  Spider-Man: No Way Home    5083.954      8940    popular
2      2021  Spider-Man: No Way Home    5083.954      8940    popular
3      2022      The Batman    3827.658      1151    popular
4      2022      The Batman    3827.658      1151    popular

      Genre
0      Action
1  Adventure
```

```
2 Science Fiction
3 Crime
4 Mystery
```

1 Data visualization

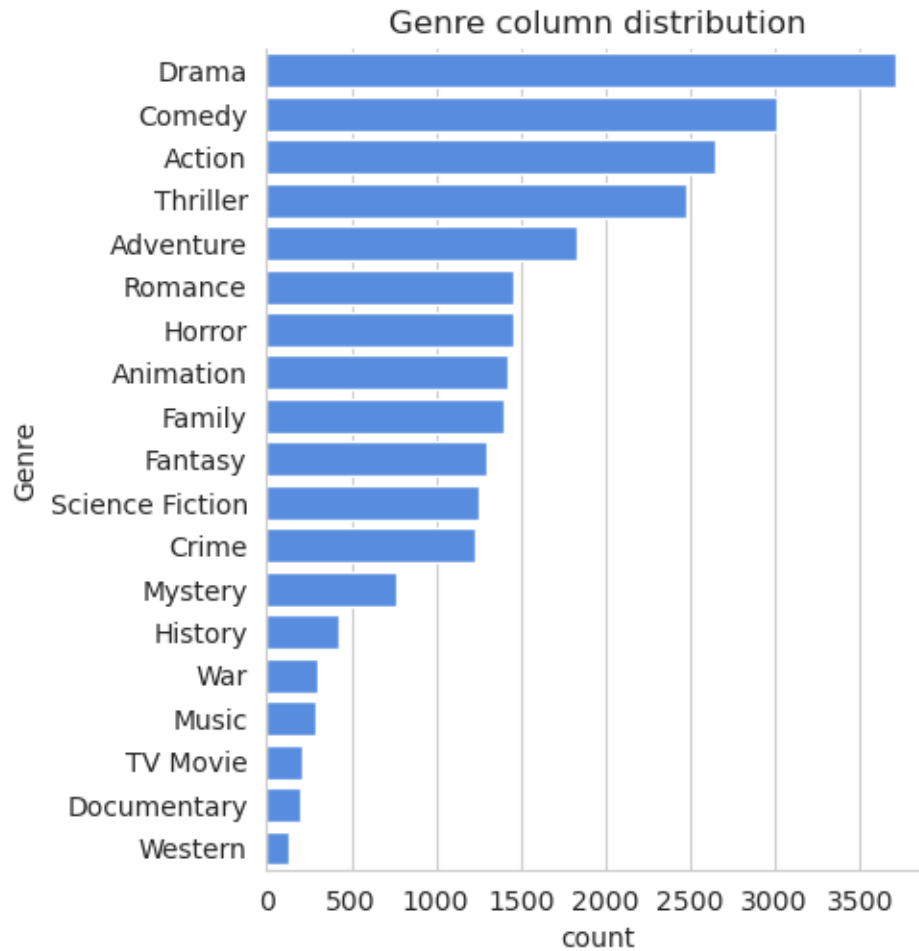
```
[28]: sns.set_style('whitegrid')
```

2 What is the most frequent genre of movies released on Netflix?

```
[29]: df['Genre'].describe()
```

```
[29]: count      25552
      unique        19
      top      Drama
      freq      3715
      Name: Genre, dtype: object
```

```
[30]: sns.catplot(y = 'Genre', data = df, kind = 'count',
                  order = df['Genre'].value_counts().index,
                  color = '#4287f5')
plt.title('Genre column distribution')
plt.show()
```



3 Which has highest votes in vote avg column?

```
[34]: df.head()
```

```
[34]:
```

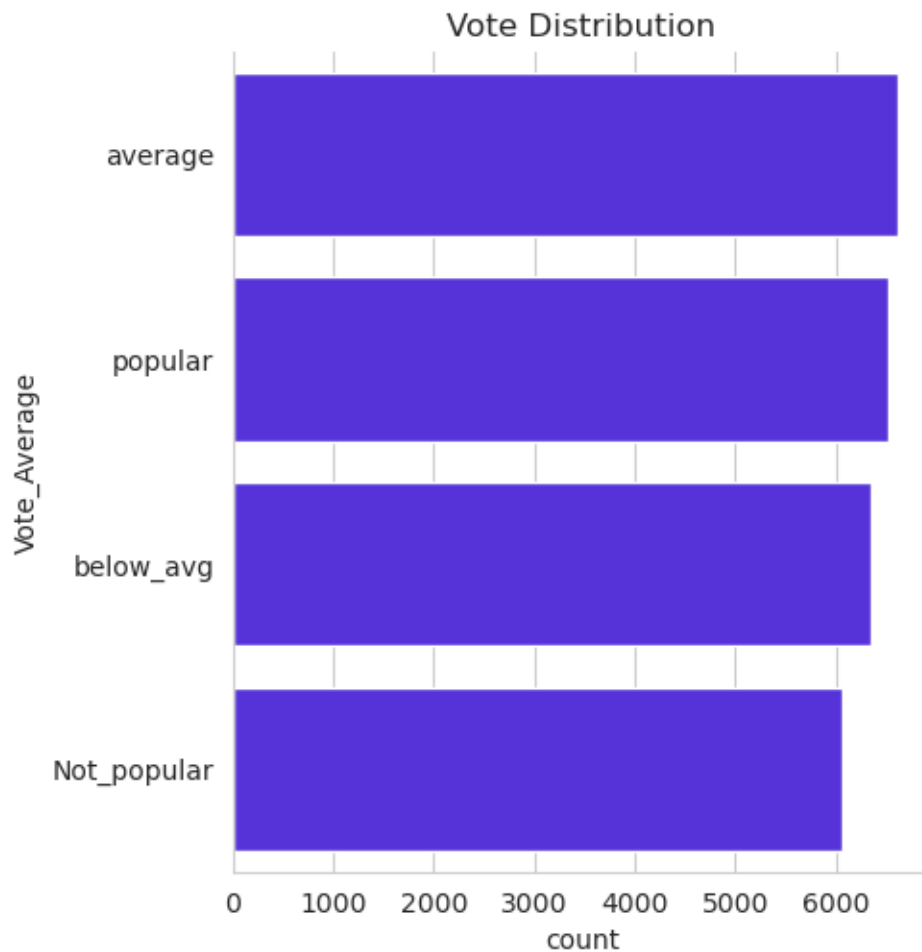
	Release_Date	Title	Popularity	Vote_Count	Vote_Average \
0	2021	Spider-Man: No Way Home	5083.954	8940	popular
1	2021	Spider-Man: No Way Home	5083.954	8940	popular
2	2021	Spider-Man: No Way Home	5083.954	8940	popular
3	2022	The Batman	3827.658	1151	popular
4	2022	The Batman	3827.658	1151	popular


```

Genre
0      Action
1  Adventure
2 Science Fiction
3      Crime

```

```
[37]: sns.catplot(y = 'Vote_Average', data = df, kind = 'count',
                order = df['Vote_Average'].value_counts().index,
                color = '#4517f5')
plt.title('Vote Distribution')
plt.show()
```



4 What movie got the highest popularity? what's its genre?

```
[38]: df.head()
```

```
[38]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	\
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	

3	2022	The Batman	3827.658	1151	popular
4	2022	The Batman	3827.658	1151	popular

	Genre
0	Action
1	Adventure
2	Science Fiction
3	Crime
4	Mystery

```
[40]: df[df['Popularity'] == df['Popularity'].max()]
```

```
[40]: Release_Date      Title  Popularity  Vote_Count  Vote_Average \
0      2021  Spider-Man: No Way Home    5083.954      8940      popular
1      2021  Spider-Man: No Way Home    5083.954      8940      popular
2      2021  Spider-Man: No Way Home    5083.954      8940      popular
```

	Genre
0	Action
1	Adventure
2	Science Fiction

5 What movie got the lowest popularity? what's its genre?

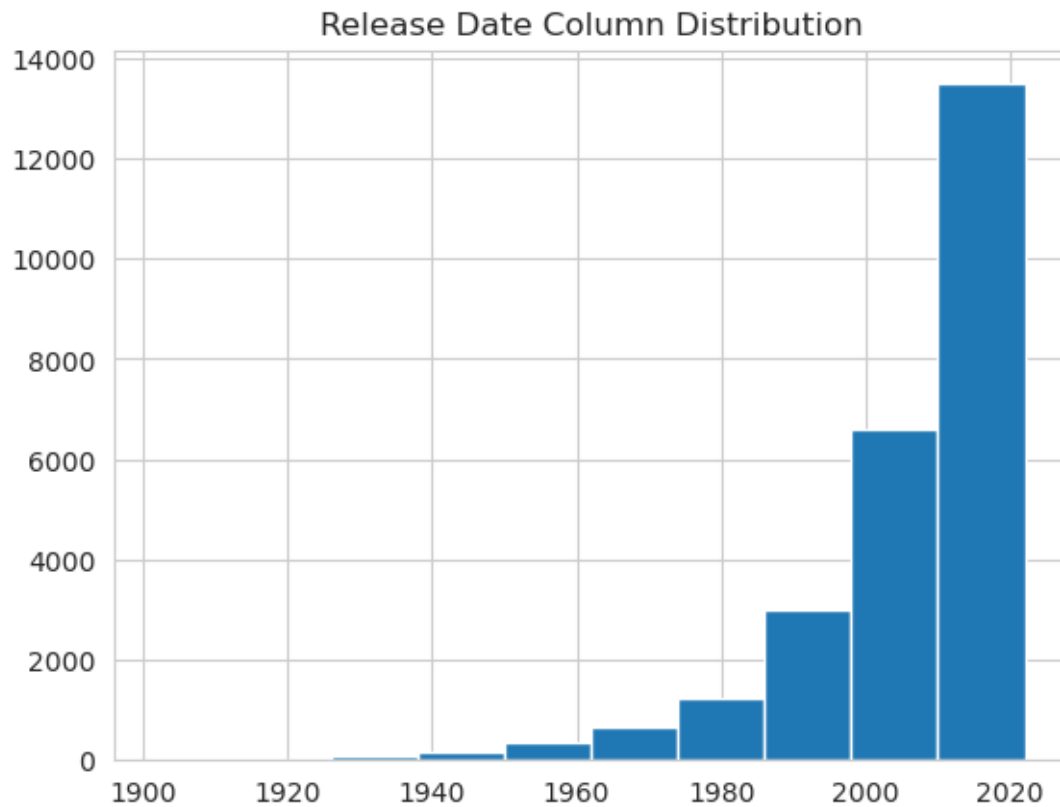
```
[41]: df[df['Popularity'] == df['Popularity'].min()]
```

```
[41]: Release_Date      Title  Popularity \
25546      2021  The United States vs. Billie Holiday    13.354
25547      2021  The United States vs. Billie Holiday    13.354
25548      2021  The United States vs. Billie Holiday    13.354
25549      1984          Threads    13.354
25550      1984          Threads    13.354
25551      1984          Threads    13.354
```

	Vote_Count	Vote_Average	Genre
25546	152	average	Music
25547	152	average	Drama
25548	152	average	History
25549	186	popular	War
25550	186	popular	Drama
25551	186	popular	Science Fiction

6 Which year has the most filmed movies?

```
[42]: df['Release_Date'].hist()  
plt.title('Release Date Column Distribution')  
plt.show()
```



```
[ ]:
```